Skills Summary

Scatter Plots, Association, and Causal Claims

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a scatter plot

How to read a scatter plot: each dot is one case, plotted as (horizontal quantity, vertical quantity).

- **Positive association:** as the horizontal value rises, the vertical value tends to rise. **Negative:** it tends to fall.
- To say how big the change is, subtract the two **vertical** values — change = end value − start value.
- Subtracting the horizontal values answers a different question. Ask yourself which axis the question is about *before* you subtract.

Example: A plot has the points $(2,5)$, $(4,8)$, $(6,10)$ and $(8,15)$. Describe the association and give the change in the vertical quantity across the plot.
Step 1. The question is about the vertical quantity, so read the vertical values at the two ends and subtract them — the horizontal values only say where to look.
Step 2. At the left end the height is 5; at the right end it is 15. Change $= 15 - 5$.
The vertical quantity rises by **10**, so the association is positive.

Try it: A plot has the points $(1,20)$, $(3,16)$, $(5,11)$ and $(7,9)$. What is the change in the vertical quantity from the left end to the right end?

check: **−11**

Skill 2 — Does the gap prove cause?

A gap between two groups is a starting point, not a verdict. Compute each group's mean from *that group's* data only, then ask three questions.

- **Lurking variable:** what else differs between these groups?
- **Reverse cause:** could the outcome have produced the grouping?
- **Who decided?** If people **chose** their own group, the study is observational and supports association only. Only **random assignment** makes the groups alike beforehand, and that is what lets a later difference be credited to the treatment.

Example: Students who joined a study club scored 90, 72, 84; students who did not scored 70, 68, 75. Find each group's mean, then judge the claim "the club raises scores".
Step 1. Compare like with like: average each group using only its own scores, because a pooled average would answer a different question.
Step 2. Add each group's three scores and divide by three.
Club mean **82**; no-club mean **71**. The club group is ahead — but the students joined the club themselves, so the gap is association, not proof of cause.

Try it: One group scored 88, 92, 84; the other scored 80, 74, 77. Find both means.

check: **88** and **77**

Skill 3 — Fixing a cause headline

Swap the causal verb for a "goes with" verb, and say who the data covers.

- Out: *causes*, *makes*, *leads to*, *raises*.
- In: *is associated with*, *tends to go with*, *students who ... also ...*
- Scope it to the data actually collected ("among these six students"), never to everyone.
- Then say what evidence *would* settle it: assign the treatment at random and compare.

Example: A plot of screen hours against sleep hours holds $(0, 9)$, $(2, 8)$, $(4, 7)$ and $(6, 6)$. Fix the headline "screens steal sleep".
Step 1. First measure what the plot really reports — the change in the vertical quantity, sleep, across the range of screen time.
Step 2. From 0 hours of screen time to 6 hours, sleep goes from 9 down to 6: change $= 6 - 9$.
Sleep changes by $\mathbf{-3}$ hours, so the honest headline is "in this group, more screen time went with less sleep".

Try it: A plot holds $(0, 60)$, $(1, 54)$, $(2, 47)$ and $(3, 39)$. What is the change in the vertical quantity across the plot?

check: $\mathbf{-21}$

(!) Watch out: A strong pattern is not strong evidence of cause. The tighter the dots sit around a line, the more tempting the cause story gets — but tightness measures how well one quantity *predicts* the other, and prediction is not causation.